



Extracytoplasmic sigma factor AlgU contributes to fitness of *Pseudomonas aeruginosa* PGPR2 during corn root colonization

Ramamoorthy Sivakumar¹ · Paramasamy Gunasekaran² · Jeyaprakash Rajendhran¹

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Abstract

In bacteria, sigma factors are crucial in determining the plasticity of core RNA polymerase (RNAP) while promoter recognition during transcription initiation. This process is modulated through an intricate regulatory network in response to environmental cues. Previously, an extracytoplasmic function (ECF) sigma factor, AlgU, was identified to positively influence the fitness of *Pseudomonas aeruginosa* PGPR2 during corn root colonization. In this study, we report that the inactivation of the *algU* gene encoded by PGPR2_23995 hampers the root colonization ability of PGPR2. An insertion mutant in the *algU* gene was constructed by allele exchange mutagenesis. The mutant strains displayed threefold decreased root colonization efficiency compared with the wild-type strain when inoculated individually and in the competition assay. The mutant strain was more sensitive to osmotic and antibiotic stresses and showed higher resistance to oxidative stress. On the other hand, the mutant strain showed increased biofilm formation on the abiotic surface, and the expression of the *pelB* and *pslA* genes involved in the biofilm matrix formation were up-regulated. In contrast, the expression of *algD*, responsible for alginate production, was significantly down-regulated in the mutant strain, which is directly regulated by the AlgU sigma factor. The mutant strain also displayed altered motility. The expression of RNA binding protein RsmA was also impeded in the mutant strain. Further, the transcript levels of genes associated with the type III secretion system (T3SS) were analyzed, which revealed a significant down-regulation in the mutant strain. These results collectively provide evidence for the regulatory role of the AlgU sigma factor in modulating gene expression during root colonization.

Keywords Extracytoplasmic sigma factor · Root colonization · Biofilm · Stress response · Motility · Type III secretion system

Introduction

Bacteria of the genus *Pseudomonas* are well-known for their efficient colonization ability in multiple habitats and metabolic versatility. Regulation of gene expression in this bacterium is a highly intricate process orchestrated by dynamic regulatory networks, which makes these bacteria adapt and survive in varied niches. For example, the genome of *P. aeruginosa* PAO1 encompasses 550 ORFs coding for transcriptional regulators, constituting 10% of the genes of

this organism (Stover et al. 2000; Potvin et al. 2008). The genes encoding these regulatory factors include 120 genes for the two-component system (TCS) and 24 genes for sigma factors with known or putative functions. Of these, 19 to 21 sigma factors belong to the alternative or extracytoplasmic function (ECF) subfamily (Chevalier et al. 2019). Similarly, the genome of *P. putida* KT2440 comprises 450 open reading frames (ORFs) coding for transcriptional regulators (Nelson et al. 2002) in contrast to *Escherichia coli*, which consists of only 266 ORFs (Martínez-Bueno et al. 2002).

Sigma factors are multidomain subunits and an integral part of bacterial RNA polymerase (RNAP), forming the holoenzyme by binding to the core enzyme. The sigma factors play significant roles in promoter recognition and transcription initiation. In addition to the primary sigma factor $\sigma 70$, which carries out the bulk of transcription, bacteria modulate the gene expression by modifying the promoter specificity of RNAP by engaging the alternate sigma factors. Of these, the

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✉ Jeyaprakash Rajendhran
jrajendhran@gmail.com

¹ Department of Genetics, School of Biological Sciences, Madurai Kamaraj University, Madurai 625 021, India

² VIT Bhopal University, Sehore, India

Extra Cytoplasmic Function sigma factor (ECF σ) comprises the dominant group identified by homology searches with primary sigma factor $\sigma 70$ (Paget and Helmann 2003). Bacterial genomes, on average, contain six ECF σ s, sequestered by anti-sigma factors and maintained inactive (Staroń et al. 2009; Helmann 2002). In response to specific environmental stimuli, the regulatory network aids in the release of bound ECF σ , recruits RNAP, and triggers the expression of target genes. In our previous study, we employed a forward genetic screen, transposon insertion sequencing (INSeq), and identified the fitness genes essential for root colonization in *P. aeruginosa* PGPR2 (Sivakumar et al. 2019). Transposon insertion in the ECF σ *algU* gene impeded root colonization by the PGPR2 strain.

AlgU (also known as AlgT, $\sigma 22$, and σE) is the sigma factor that regulates the transition from nonmucoid to mucoid state in *P. aeruginosa* and modulates gene expression in response to cell wall stress (Lyczak et al. 2002; Wood and Ohman 2009). AlgU is a 22.2 kDa protein that regulates the expression of genes implicated in alginate biosynthesis. The alginate biosynthesis gene cluster encompasses 12 genes forming the *algD* operon (Ramsey and Wozniak 2005). The expression of the *algU* gene is tightly controlled by the anti-sigma factor MucA (Schurr et al. 1996; Xie et al. 1996). The *mucA* gene codes for a 21.0 kDa protein that lies downstream of the *algU* sigma factor, which is part of the *algU-mucABCD* operon (Schurr et al. 1994). The cytoplasmic domain of MucA binds the AlgU inhibiting the association of the σ factor with RNAP to form the holoenzyme and mediate the expression of AlgU-dependant promoters. In addition to *mucA*, the expression of *algU* is negatively regulated by another periplasmic protein, MucB (Martin et al. 1993).

Regulation of *algU* expression is a dynamic and coordinated action of multiple proteins. Under the normal state, MucB binds to the C-terminal region of MucA, preventing the proteolysis of MucA by the periplasmic protease AlgW, maintaining the *algU* gene in an inactive state. In the presence of an appropriate signal, expression of the *algU* gene is induced, triggering an intramembrane proteolysis (RIP) cascade, eventually activating genes under the *algU* regulon (Wood and Ohman 2009). Accumulation of RIP initiating signal molecules such as the envelope protein MucE, truncated type IV pilin, unassembled lipopolysaccharides, unassembled OMPs, and CupB5 in the periplasm trigger proteolysis of MucA by AlgW (Damron and Goldberg 2012; Wood et al. 2006; Qiu et al. 2007). Subsequently, complete cleavage of MucA is accomplished by further proteolysis by MucP and ClpXP proteases localized at the inner membrane and cytoplasm. RIP cascades are antagonized by the MucD protease, which is part of the *algU-mucABCD* operon, by degrading RIP activation signals and thereby regulating AlgW activation.

Homologs of the *algU* gene are found in diverse bacterial lineages. Mounting evidence substantiates the importance of the AlgU sigma factor during plant–pathogen interactions. For example, in *Xanthomonas campestris* pv. *campestris* and *P. syringae*, AlgU induce the expression of type III secretion system (T3SS) and type III effector (T3E) genes and contributes significantly to the virulence of the bacterium (Bordes et al. 2011; Cheng et al. 2008; Yang et al. 2018). Similarly, in *Xylella fastidiosa* AlgU sigma factor was induced in the presence of xylem fluid and contributes to biofilm formation, facilitating heat stress tolerance and virulence in grapevines (daSilva Neto et al. 2007; Shi et al. 2007). Extensive studies have been conducted to decipher the role of AlgU in regulating the expression of virulence genes in plant-pathogenic bacteria. However, little is known about the role of the AlgU sigma factor in plant-beneficial bacteria. Here, we elucidate the role of the AlgU sigma factor in the biocontrol and growth-promoting strain of *P. aeruginosa* PGPR2 during plant root colonization. The *algU* mutant was severely hampered in its ability to colonize the corn roots individually and in a competition assay. Our results provide insights into the regulatory role of the AlgU sigma factor in the PGPR2 strain.

Materials and methods

Bacterial strains and growth conditions

The bacterial strains and plasmids used in this study are listed in Table 1. *E. coli* cultures were subcultured in Luria Bertani (LB) medium and grown at 37 °C with agitation (200 rpm). *P. aeruginosa* PGPR2 strains were grown in LB medium or M9 medium supplemented with 0.4% glucose as the sole carbon and nitrogen source (Sambrook et al. 1989) at 37 °C with or without shaking. When required, the medium was solidified using 2% agar. Antibiotics were added as needed at the following concentration (unless otherwise specified): ampicillin, 100 $\mu\text{g ml}^{-1}$; gentamicin, 15 $\mu\text{g ml}^{-1}$ for *E. coli* and 30 $\mu\text{g ml}^{-1}$ for PGPR2; irgasan, 25 $\mu\text{g ml}^{-1}$; tetracycline, 10 $\mu\text{g ml}^{-1}$ for *E. coli* and 80 $\mu\text{g ml}^{-1}$ for PGPR2.

Collection of root exudate

Root exudate was collected as described previously with some modifications (Shidore et al. 2012). Corn seeds were surface sterilized, as mentioned earlier (Sivakumar et al. 2019), and germinated in the dark for 3 days. Under sterile conditions, 10 seedlings were transferred to a container with a 30 ml plant nutrient solution. The seedlings were placed on sponge foams aseptically submerged to ensure the grain rested on the sponge, the roots dipped in the plant nutrient

Table 1 Strains and plasmids used in this study

Bacterial strains/Plasmids	Description ^a	Source or Reference
<i>P. aeruginosa</i>		
PGPR2	<i>P. aeruginosa</i> wild type strain	Illakkiam et al. (2013)
$\Delta algU$	<i>algU</i> [−] derivative of PGPR2	This study
$\Delta algU/pUCP26$	$\Delta algU$ harboring pUCP26 plasmid with no insert	This study
$\Delta algU$ – <i>algU</i>	$\Delta algU$ harboring recombinant construct (pAlgU) for complementation	This study
<i>E. coli</i>		
DH5 α	F– $\emptyset$ 80lacZ M15 <i>endA recA hsdR</i> (rk–mk–) <i>supE thi gyrA relA</i> Δ (lacZYA– <i>argF</i>) U169	Laboratory stock
S17-1 λ –pir	<i>recA pro hsdR</i> RP4-2-Tc::Mu-Km::Tn7 λ –pir	de Lorenzo et al. (1990)
Plasmids		
pIVETP	Tc ^R , Ap ^R , <i>ori</i> R6K, gene replacement vector	Rainey (1999)
pGEM-T	Ap ^R , PCR product cloning vector	Promega
pUCP24	Gm ^R , <i>E. coli</i> – <i>Pseudomonas</i> Shuttle vector	West et al. (1994)
pUCP26	Tc ^R , <i>E. coli</i> – <i>Pseudomonas</i> Shuttle vector	West et al. (1994)
pAlgU	Tc ^R , 650 bp wild type copy of <i>algU</i> gene cloned in pUCP26	This study
pAlgU1	Ap ^R , 300-bp deletion construct of <i>algU</i> gene in pGEM-T	This study
pAlgU2	Ap ^R , Gm ^R , Insertion of gentamicin cassette into 300-bp deletion construct of <i>algU</i> gene in pAlgU1 [$\Delta algU$::Gm ^R]	This study
pAlgU3	Tc ^R , Ap ^R , Gm ^R , 1.1-kb deletion construct of <i>algU</i> gene subcloned into <i>XhoI</i> site of pIVETP	This study

^aAp^R, Tc^R, and Gm^R resistance to ampicillin, tetracycline, and gentamicin, respectively

medium, and the shoots exposed to light. The bottles were maintained in greenhouse conditions of 16 h light and 8 h dark at 28 °C for 7 days. The plants were removed after 7 days, and the medium containing the exudate was filtered through a 0.45-micron filter to remove larger particles. The sterility of the exudate was determined by plating 100 μ l of exudate on a rich medium.

Quantification of *algU* expression

Reverse transcriptase quantitative PCR (RT-qPCR) was performed to quantify the expression pattern of the *algU* gene during corn root colonization. The transcript level of *algU* gene was quantified post exposure to corn root exudate. Overnight culture of PGPR2 strain was pelleted and washed three times with sterile 10 mM MgSO₄. The suspension was diluted (OD₆₀₀ = 0.2) and inoculated to a flask containing 25 ml of M9 medium supplemented with 0.4% glucose (control) and corn root exudate (test). The flasks were incubated for 6 h, and total RNA was extracted. One μ g of DNA-free RNA was converted to cDNA using the RevertAid first-strand cDNA synthesis kit (ThermoScientific, USA) following the manufacturer's instructions. Fold change was calculated by the $\Delta\Delta$ Ct method using the threshold cycle (Ct) value by comparing it with the housekeeping gene (*rpoD*) (Livak and Schmittgen 2001). The experiment was performed in triplicate.

Construction of the *algU* mutant

The *algU* mutant strain was constructed by inserting an antibiotic cassette in-between the coding region of the *algU* gene. PCR primers were designed containing *HindIII* restriction sites to amplify 150 bp fragment of the *algU* gene at the 5' and 3' regions (Table S1). Each of the 150 bp PCR products was digested with *HindIII* and ligated. The ligation mix was cleaned up and used as a template to amplify the 300 bp ligated product using primers AlgU FP and AlgU RP (Table S1). The 300 bp amplicon was cloned in the pGEM-T easy vector yielding the plasmid pAlgU1 (Table 1). The gentamicin-resistance cassette was amplified from the pUCP24 vector (Table 1) with primers Gen FP and Gen RP containing *HindIII* restriction sites (Table S1). The gentamicin-resistance cassette and pAlgU1 plasmid was restricted with *HindIII*, eluted, ligated, and transformed into *E. coli* DH5 α resulting in plasmid pAlgU2. The in vitro deleted product ($\Delta algU$::Gm^R) was then cloned into the *XhoI* site of suicide vector pIVETP yielding the plasmid pAlgU3 (Table 1), which was mobilized into *P. aeruginosa* PGPR2 from *E. coli* S17-1 λ –pir. The exconjugants were selected on LB agar containing gentamicin and irgasan. The plasmid loss was verified by PCR amplification of the tetracycline resistance gene. The mutant was confirmed by PCR and sequencing. The workflow followed for mutant construction is shown in Fig. S1. Growth analysis was

performed between wild type and the mutant strain using LB medium to determine any growth deficiency in PGPR2 due to *algU* mutation.

Construction of complementation strain

The *Pseudomonas*–*E. coli* shuttle vector pUCP26 was used to construct a recombinant plasmid to complement the function of the *algU* gene *in trans* in the mutant strain. Using genomic DNA from PGPR2 as the template, the wild-type copy of the *algU* gene along with its native promoter was amplified using primers *algU* comp FP and *algU* comp RP (Table S1). The PCR products were restricted with *Hind*III enzyme and cloned in *Hind*III site of pUCP26 plasmid resulting in pAlgU construct. Electrocompetent cells of Δ *algU* were prepared using ice-cold 10% glycerol. The plasmids pAlgU and pUCP26 with no insert were transformed in to Δ *algU* strain by electroporation using Gene Pulser (Biorad). The transformants were plated on LB agar mediums supplemented with tetracycline (80 μ g ml^{−1}). The Δ *algU* strain harboring the recombinant construct was confirmed by PCR.

Root colonization assay

Root colonization and competitive root colonization assays were performed as previously described (Sivakumar et al. 2019). The germinated corn seedlings were transferred aseptically to a gnotobiotic hydroponic medium (one seedling per 50 ml of plant nutrient solution) and maintained under greenhouse conditions (16 h light and 8 h dark at 28 °C). Corn plantlets grown in hydroponic systems were inoculated with $\sim 10^6$ cells of the wild type, the mutant, and the complementation strain. At 7 days post inoculation, the roots were aseptically excised and transferred to a 50 ml tube containing 20 ml 0.85% saline and ten glass beads (5 mm). The tube was vortexed for 1 min to detach adhered bacteria, and the suspension was enumerated by standard dilution plating. For competitive root colonization assay, the wild type and the mutant strain were diluted to $\sim 10^6$ cells in 10 mM MgSO₄, and a 1:1 mixture of both strains were used as inoculum. Each experiment was repeated three times, with five replicates per treatment.

Copy number determination using quantitative real-time PCR

To determine the copy number of the mutant and the wild-type strain during competitive root colonization, we performed absolute quantification by real-time PCR. A competitive root colonization assay was performed as described previously (Sivakumar et al. 2019). At 7 days post inoculation, the roots were aseptically excised and

transferred to a 50 ml tube containing 20 ml 0.85% saline and ten glass beads. The tubes were vortexed for 1 min to detach adhered bacteria, and the suspension was used to isolate total genomic DNA (g-DNA). The g-DNA was used as the template to perform qPCR at a final concentration of 1 ng μ l^{−1}. Varied concentrations (50 ng–5 μ g) of g-DNA from PGPR2 were used as the template to amplify the V3 and V4 region of 16S rRNA using primers 331F and 531R, which served as standard. A standard curve was plotted from the Ct values; the slope and y-intercept were calculated. The primers Gen qPCR FP, Gen qPCR RP, AlgU qPCR FP, and AlgU qPCR RP (Table S1) were used to amplify the mutant-specific and the wild-type-specific genes.

$$E_{AMP} = 10^{-1/\text{Standard curve slope}}$$

$$\% \text{Efficiency} = (E_{AMP} - 1) \times 100$$

$$\text{No. of copies} = E_{AMP}^{\text{Standard curve intercept}-\text{Ct value}}$$

Stress susceptibility assay

The wild type and the mutant strain were assayed for the ability to resist oxidative, osmotic, and antibiotic stress. To induce oxidative stress, exponential-phase cultures of the wild type, the mutant and the complementation strains were diluted to the same density (OD₆₀₀=0.6) and treated with 4 mM H₂O₂ for 30 min. The cultures were serially diluted in fresh LB, plated on prewarmed LB medium and incubated at 37 °C. The ability of the wild type and the mutant strain to resist oxidative stress was analyzed by enumerating the colony-forming units. Similarly, the ability of the *algU* mutant strain to resist osmotic stress was measured by serially diluting the wild type and the mutant strain (OD₆₀₀=0.6) and plating on LB medium containing 0.8 M NaCl. To assess the ability of the *algU* mutant strain to resist antibiotic stress, disc diffusion assay was performed using commercial antibiotic discs of imipenem and polymyxin B (HiMedia). Exponential-phase cultures of the wild type, the mutant, and the complementation strains were spread on Mueller Hinton Agar using a sterile cotton swab. Each antibiotic disc was placed on the center of plate and incubated at 37 °C. The plates were observed for the zone of clearance to measure the degree of susceptibility.

Reverse transcriptase quantitative polymerase chain reaction (RT-qPCR)

The transcript levels of *rsmA* and genes associated with T3SS (*exsD*, *exsC*, and *popB*) were measured in the wild type and the mutant strain post exposure with root exudate.

The wild type and the mutant strains were exposed to corn root exudate for 8 h, cells were harvested, and total RNA was harvested. One μg of DNA-free RNA was converted to cDNA using the RevertAid first-strand cDNA synthesis kit (ThermoScientific, USA) following the manufacturer's instructions. Fold change was calculated by the $\Delta\Delta\text{Ct}$ method using the threshold cycle (Ct) value by comparing it with the housekeeping gene (*rpoD*) (Livak and Schmittgen 2001).

Motility assay

Swimming motility assay was performed in a polystyrene Petri dish (90 mm diameter) as described previously with modification (Murray and Kazmierczak 2006). Briefly, 5 μl of overnight grown cultures of the wild type, mutant, and complementation strains were stabbed directly into the center of 0.2% minimal medium agar plate. The swimming zone was measured overnight post incubation at 37 °C for 24 h. The experiment was repeated three times independently in triplicates.

The subsurface twitching motility assay was carried out as described previously with modifications (Semmler et al. 1999). The overnight grown cultures of the wild type, the mutant, and the complementation strains were stab inoculated through a 1% agar plate and incubated overnight at 37 °C. The zone of motility between the agar and the petri dish interface was visualized by compressing the agar and then staining using 0.05% (wt/vol) Coomassie brilliant blue R250 stain (40% methanol, 10% acetic acid) as described previously (Alm and Mattick 1995). The transcript levels of *fliC*, *flhF*, and *pilZ* were measured in the wild type and the mutant strain post exposure with root exudate as mentioned above.

Biofilm formation

Biofilm formation assay was performed as described previously with some modifications (Müsken et al. 2010). Overnight cultures of the wild type, mutant, and complementation strains were diluted to 1:1000 (vol/vol) in fresh LB medium. 1 ml of the diluted cultures were seeded into each well of a 24-well plate. The sterile medium served as the negative control. The plates were incubated at 37 °C in a humidified chamber. After 48 h, mature biofilms were washed thrice with sterile saline, air dried, stained with crystal violet (0.5% wt/vol) for 20 min, washed thrice with sterile saline, and air dried. The crystal violet bound to wells was solubilized with 95% ethanol, and absorbance was measured at 575 nm in a plate reader. Biofilm formation was considered significantly higher between the wild

type and the mutant strain when the p value <0.05 . The transcript levels of genes associated with biofilm formation (*algD*, *pelB*, *pslJ*, and *cup* fimbriae) were analyzed in the wild type and mutant strains. Biofilm of the wild type and the mutant strain was grown as mentioned above, and total RNA was extracted post 48 h of incubation and the expression profile was analyzed as mentioned above.

Biofilm-forming abilities of the mutant, the wild type and, complementation strains were further validated using fluorescein diacetate (FDA) assay. The FDA assay was performed as described previously with modifications (Peeters et al. 2008). Briefly, biofilm was grown as described above for 24 h. The planktonic cells were carefully removed and washed thrice with sterile 3-(N-Morpholino) propane sulfonic acid (MOPS) buffer (20.9 g MOPS, 5.6 g NaCl for 1 L at pH 7.0). To each well, 1 ml of 10 mg/ml FDA diluted at a 1:50 ratio in MOPS buffer was added and incubated for 2 h in the dark. After incubation, the FDA solution was transferred into a new optiplate-96F (Part No: 6005270, Perkin Elmer, USA), and fluorescence was measured (Excitation: 494 nm and emission: 518 nm) in Enspire multimode plate reader (Perkin Elmer, USA). The relative fluorescent units (RFU) between the wild type, the mutant and the complementation strains were measured, and considered significant when the p value <0.05 .

Exopolysaccharide (EPS) quantification

The ability of the wild type, mutant, and complementation strains to produce EPS was quantified as previously described with modifications (Nithya et al. 2011). Briefly, biofilms of the wild type, mutant and complementation strains were grown for 48 h as described above. The planktonic cells were carefully aspirated, and sessile cells were washed thoroughly using sterile saline (0.9% NaCl) and resuspended in 500 μl of saline. The cell suspensions were transferred to sterile culture tubes. To this, an equal volume of 5% of phenol and five volumes of concentrated sulphuric acid containing 0.2% hydrazine sulfate was added and mixed gently and incubated in the dark for 1 h. Optical density was measured at 490 nm. The experiment was done three times, and the mean values were represented.

Statistical analysis

All experiments were performed three times, the standard deviation was calculated, and statistical significance was determined using a two-tailed Student's t test.

Results

Inactivation of AlgU sigma factor hampers root colonization in PGPR2

Reverse transcriptase quantitative PCR revealed a significant increase in the expression of the *algU* gene. Expression of the *algU* gene was up-regulated by 2-fold ($p = 0.047$) in PGPR2 post exposure to root exudate (Fig. 1a). Root exudate serve as a model for understanding the gene expression pattern in the plant–microbe interactions and provide an alternative to overcome the limitations of in vivo transcriptomics during plant–symbiont interactions owing to the imbalance of biomass ratio (Fan et al. 2012; Zhang et al. 2015; Shidore et al. 2012; Yi

et al. 2017). To understand the molecular mechanism, an *algU* mutant of *P. aeruginosa* PGPR2 was constructed by antibiotic-mediated insertional inactivation. Mutation in the *algU* gene was confirmed by PCR and sequencing (Fig. S2). Growth analysis between the wild type and the mutant strain revealed that the growth was unaffected under in vitro conditions (Fig. S3). The mutant strain's ability to colonize corn roots was assayed, individually and in competition with the wild-type strain. When inoculated separately, a threefold reduction ($p < 0.05$) in the colonization of corn root was observed with the *algU* mutant after 7 days of incubation which was reverted to the level of wild type in the complementation strain (Fig. 1b, c). Similarly, when the wild type and the mutant strain were inoculated to the same plant (competitive assay), the wild type represented 76% of the viable bacterial cells recovered

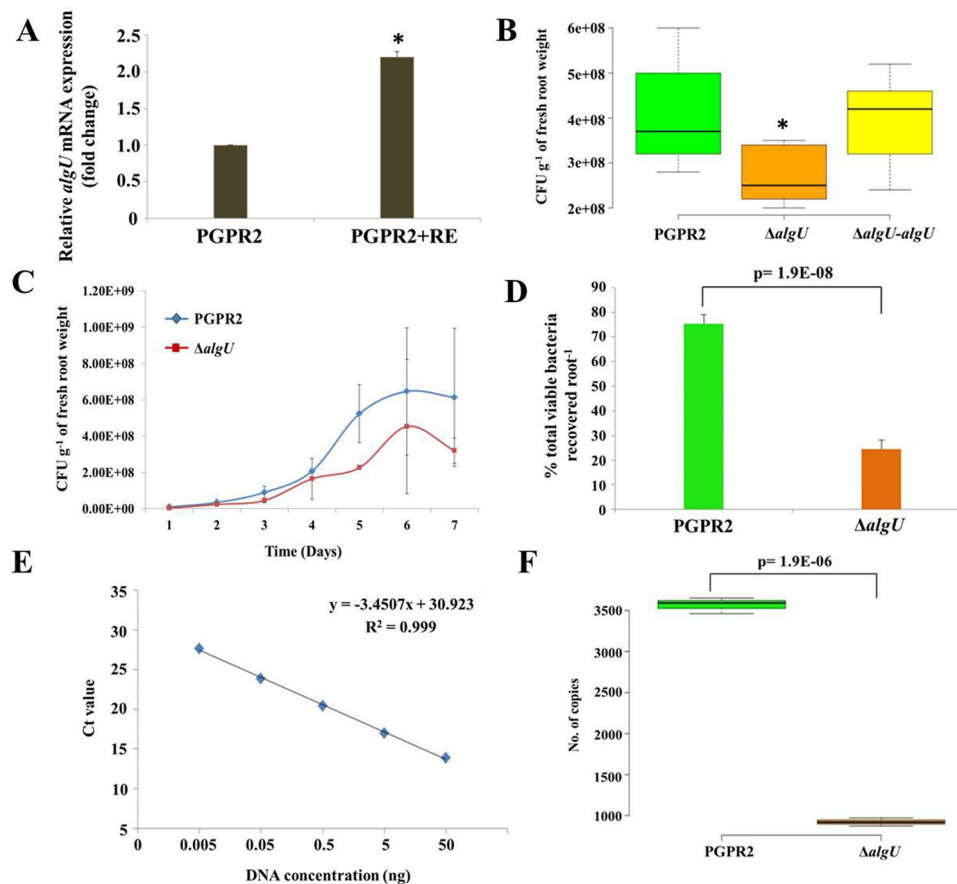


Fig. 1 Corn root colonization by *P. aeruginosa* PGPR2 and AlgU isogenic mutants. **A** Quantification of transcript level of *algU* gene in PGPR2 post exposure to corn root exudate. $*p = 0.047$. **B** 4×10^6 CFU ml⁻¹ of the PGPR2, the mutant and the complementation strain were inoculated individually, and bacterial populations were determined seven days post inoculation. $*p = 0.035$ **C** Time course growth analysis of the wild type and the mutant strain during root colonization. **D** Competitive corn root colonization by the PGPR2 and the *algU* mutant. 1:1 mixture of the wild type and the

mutant strain were inoculated; the competitive colonizing ability was expressed as a percentage of viable bacterial cells recovered from the root. The colonization experiment was performed thrice independently with five replicates within each experiment. **E** The standard curve plotted using Ct values for the 16S rDNA gene obtained from qPCR. **F** Copy number determination of the wild type and the mutant strain post competitive root colonization assay by quantitative PCR

from the root, suggesting that the mutant strain was unable to compete with the wild-type strain in this niche (Fig. 1d). This result was further validated by determining the copy number of the wild type and mutant strain using quantitative real-time PCR. A standard curve was constructed using g-DNA of PGPR2 with five serial dilutions ranging from 50 ng to 5 pg (Fig. S4). The efficiency was 98.8%, the coefficient of correlation (R^2) was 0.99, the slope (m) was -3.45 , and the y-intercept was 30.9 (Fig. 1e). The copy number of the mutant strain decreased by 3.5-fold

during competitive root colonization compared with the wild-type strain (Fig. 1f).

Mutants of *algU* display altered tolerance to environmental stress

As cell wall stress activates the *algU*-dependent genes (Wood and Ohman 2009), we analyzed the mutant and the wild-type strain's ability to resist osmotic, oxidative, and

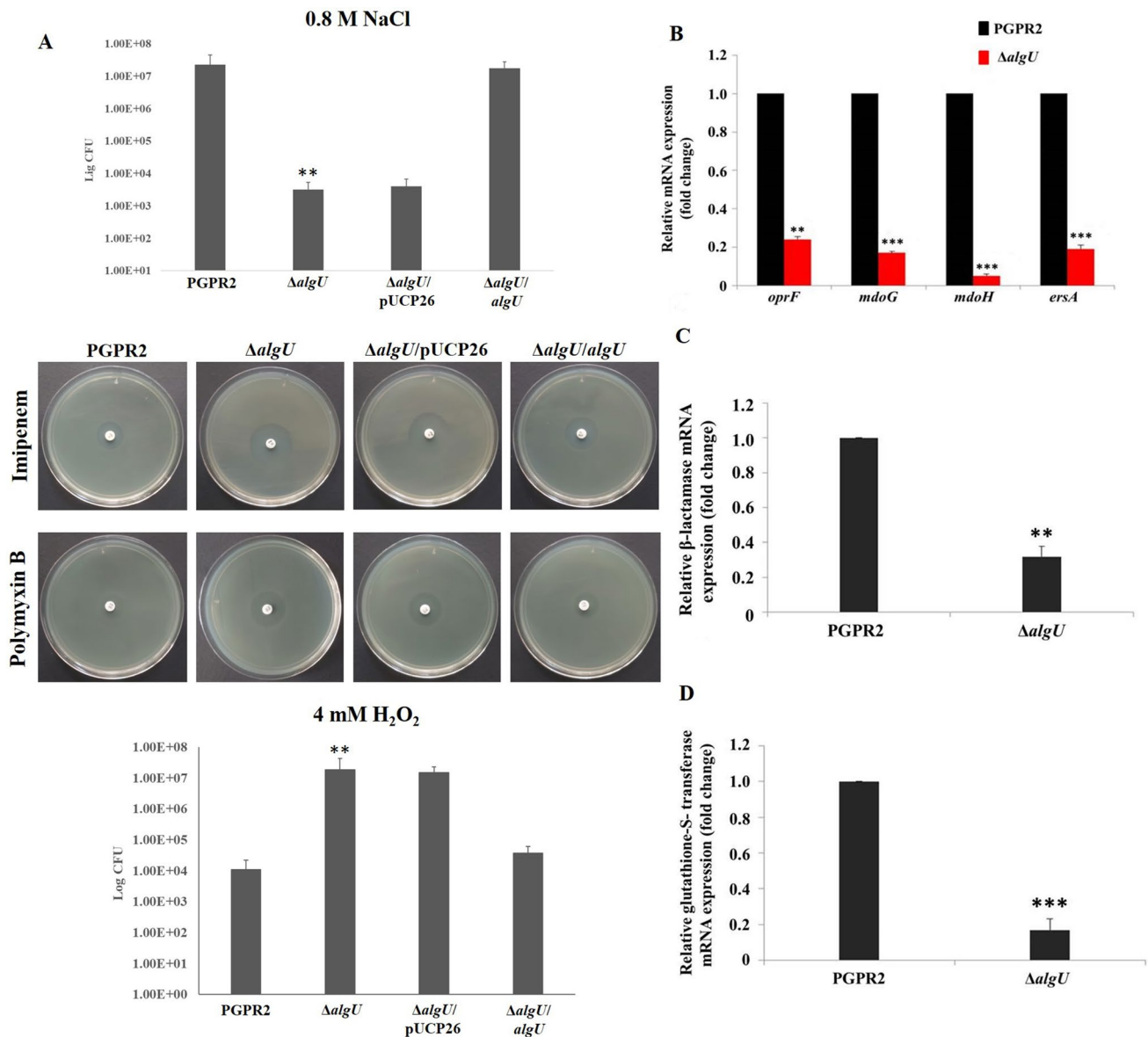


Fig. 2 Inactivation of *algU* gene alters stress tolerance. **A** The wild type, the mutant and the complementation strains were tested for their ability to resist different stressors. The colony-forming unit between the wild type and the mutant in the presence of stress inducer indicates the degree of susceptibility or resistance. Significant difference Osmotic stress $**p=0.001$; Oxidative stress $**p=0.003$. **B** Effect of *algU* mutation on the expression of genes associated with

osmotic stress resistance under *algU* regulon. Significant difference *oprF* $**p=0.00082$; *mdoG* $***p=3.2E-05$; *mdoH* $***p=2.7E-09$; *ersA* $***p=0.00017$ **C** Effect of *algU* mutation on expression of the β -lactamase gene. $**p=0.0028$ **D** Effect of *algU* mutation on the expression of glutathione-S-transferase involved in oxidative stress resistance. $***p=1.2E-05$

antibiotic stress. The growth of the mutant strain was drastically hampered under osmotic stress, which displayed a 1000-fold reduced growth in a medium containing 0.8 M NaCl, while the complementation strain showed a similar growth pattern of the wild-type strain (Fig. 2a). We quantified the transcript levels of three genes of the *algU* regulon (*oprF*, *mdoG*, and *mdoH*), which play a crucial role in osmotic stress tolerance (Firoved et al. 2002; Wood and Ohman 2009). We observe a significant down-regulation of all three genes by 4.1-, 5.8-, and 20-fold, respectively, in the mutant strain (Fig. 2b). In addition, we also measured the mRNA level of sRNA *ersA*, which plays a vital role in osmotic stress tolerance, and directly regulated by the AlgU sigma factor (Ferrara et al. 2015; Park et al. 2014). A 5.2-fold reduction in the expression level of *ersA* was observed in the mutant strain during osmotic stress induced by 0.8 M NaCl.

Similarly, the mutant strain was highly susceptible to antibiotic stress induced by imipenem and polymyxin B, evident by disc diffusion assay (Fig. 2a). This phenotype was reverted to the wild-type level in the complementation strain. Since the mutant strain displayed decreased resistance to beta-lactam antibiotic (imipenem), the transcript level of a gene coding for β -lactamase (PGPR2_22615) was measured. We observed a significant down-regulation by 3.1-folds in the mutant strain when compared with the wild type (Fig. 2c). The mutant strain displayed higher resistance to hydrogen peroxide-induced oxidative stress in comparison to the wild-type strain, which was reverted in the complementation strain (Fig. 2a). Despite the enhanced tolerance to oxidative stress, the expression of a gene coding for glutathione S-transferase (PGPR2_18440) was found to be significantly down-regulated by 5.8-fold in

the mutant strain, known to play a crucial role in scavenging reactive oxygen species (Fig. 2d).

The expression of the *rsmA* gene was down-regulated in *algU* mutants

RsmA is a global regulator of gene expression in *P. aeruginosa* (Brencic and Lory 2009). A previous study reported that, in a *mucA* mutant background, the expression of *rsmA* was increased significantly, which was positively regulated through the AlgU sigma factor (Intile et al. 2014; Stacey and Pritchett 2016). To corroborate this regulatory mechanism, we quantified the *rsmA* transcript level in the *algU* mutant background and the wild-type post exposure to root exudate. A 3.3-fold down-regulation of *rsmA* expression was observed in the *algU* mutant strain compared with the wild-type strain (Fig. 3a).

The *algU* mutation affects the expression of T3SS genes

In addition to motility, RsmA regulates genes associated with T3SS, which play a vital role during root colonization (Intile et al. 2014). We analyzed the transcript levels of three genes (*exsC*, *exsD*, and *popB*) involved in T3SS post exposure to root exudate. Of these, *exsC* and *exsD* code for regulatory proteins of T3SS, while *popB* codes for a translocator protein. In the *algU* mutant strain, the expression of all three genes was down-regulated. We observed a 9.1-fold decreased expression of the *exsC* gene and a 12-fold decreased expression of the *exsD*. The expression of the *popB* was 30-fold decreased in the *algU* mutant strain related to the wild-type post-exposure to root exudate (Fig. 3b).

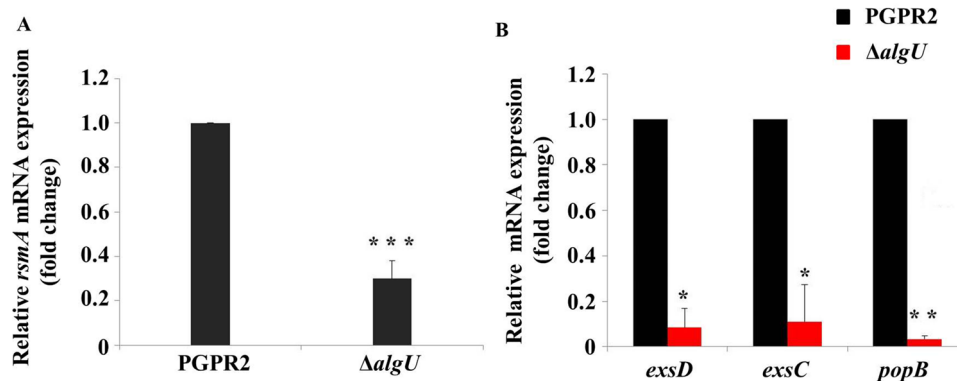


Fig. 3 **A** Effect of *algU* mutation on the expression of *rsmA* gene in the PGPR2 strain post exposure to root exudate. * $p=0.00012$ **B** Transcript profiling of T3SS genes in the *algU* mutant in response to root exudate. Significant difference *exsD* * $p=0.007$; *exsC* * $p=0.0058$; *popB* ** $p=0.0008$. The gene expression value in the wild-type strain

for these genes was considered as 1 and calculated the relative change in expression in the mutant strain. Reverse transcriptase quantitative PCR was performed in triplicate, and the data presented were mean of relative change, and the error bar represents standard deviation

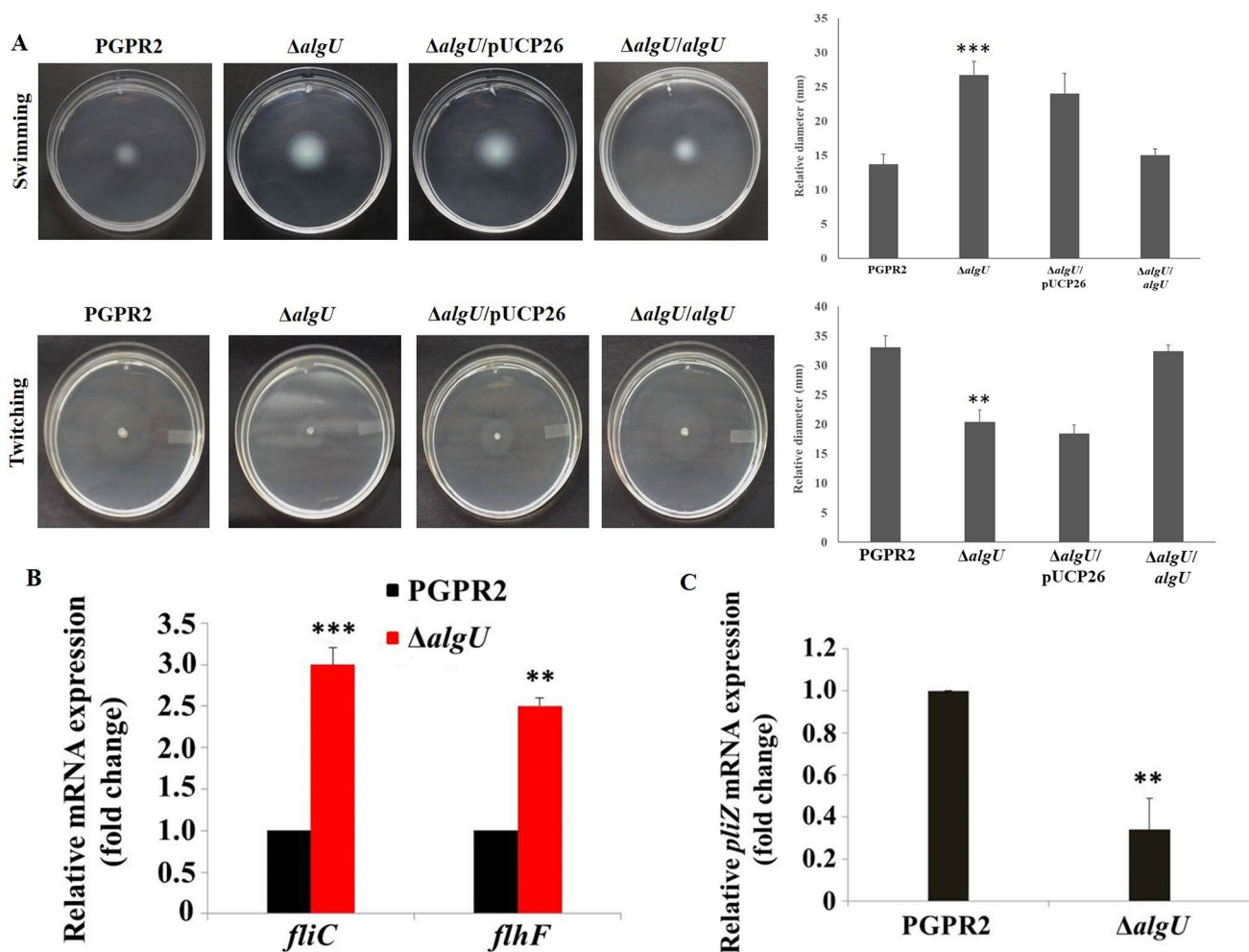


Fig. 4 Motility analysis of PGPR2 and *algU* isogenic mutants. **A** Qualitative assessment of swimming and twitching motility in minimal medium agar (0.2% and 1%) containing 0.4% glucose and incubated at 37 °C for 24 h. Swimming motility significant difference * $p=0.0009$; twitching motility significant difference * $p=0.001$. **B** Effect of *algU* mutation on the expression of *fliC* and *flhF* in the PGPR2 upon exposure to root exudate. Significant difference *fliC* *** $p=0.0002$; *flhF* ** $p=0.0032$. **C** Effect of *algU* mutation on

the expression of *pilZ* in the PGPR2 upon exposure to root exudate. Significant difference ** $p=0.004$. The gene expression value in the wild-type strain for these genes was considered as 1 and calculated the relative change in expression in the mutant strain. Reverse transcriptase quantitative PCR was performed in triplicate, and the data presented are mean of relative change, and the error bar represents the standard deviation

Inactivation of AlgU sigma factor alters motility

The post-transcriptional regulator RsmA controls the expression of various genes implicated in motility (Martinez-Granero 2012; Muriel et al. 2019). As the expression of *rsmA* was significantly down-regulated in the *algU* mutant background, we examined the ability of mutant strains to swim and twitch on 0.2% and 1% agar plates. The mutant strain's ability to swim was significantly increased compared with the wild-type strain (Fig. 4a). This result was further validated by measuring the relative transcript levels of the *fliC* gene, coding for flagellin and *flhF* gene, which plays a crucial role in swimming motility of *Pseudomonas* (Murray and Kazmierczak 2006). The expression of *fliC* and *flhF* was up-regulated

by 3.0- and 2.5-fold in the mutant strain (Fig. 4b). In contrast, type IV pili-driven twitching motility was significantly hampered in the mutant strain (Fig. 4a). This observation was consistent with our qPCR analysis, a significant down-regulation of the *pilZ* gene by 2.9-fold was observed in the mutant strain, which is vital for type IV pili biogenesis (Fig. 4c).

The *algU* mutant formed increased biofilms on abiotic surfaces

The gene coding for the post-transcriptional regulator, *rsmA* was down-regulated in the *algU* mutant, negatively modulating the expression of biofilm formation genes. In addition, the mutant strain displayed altered stress

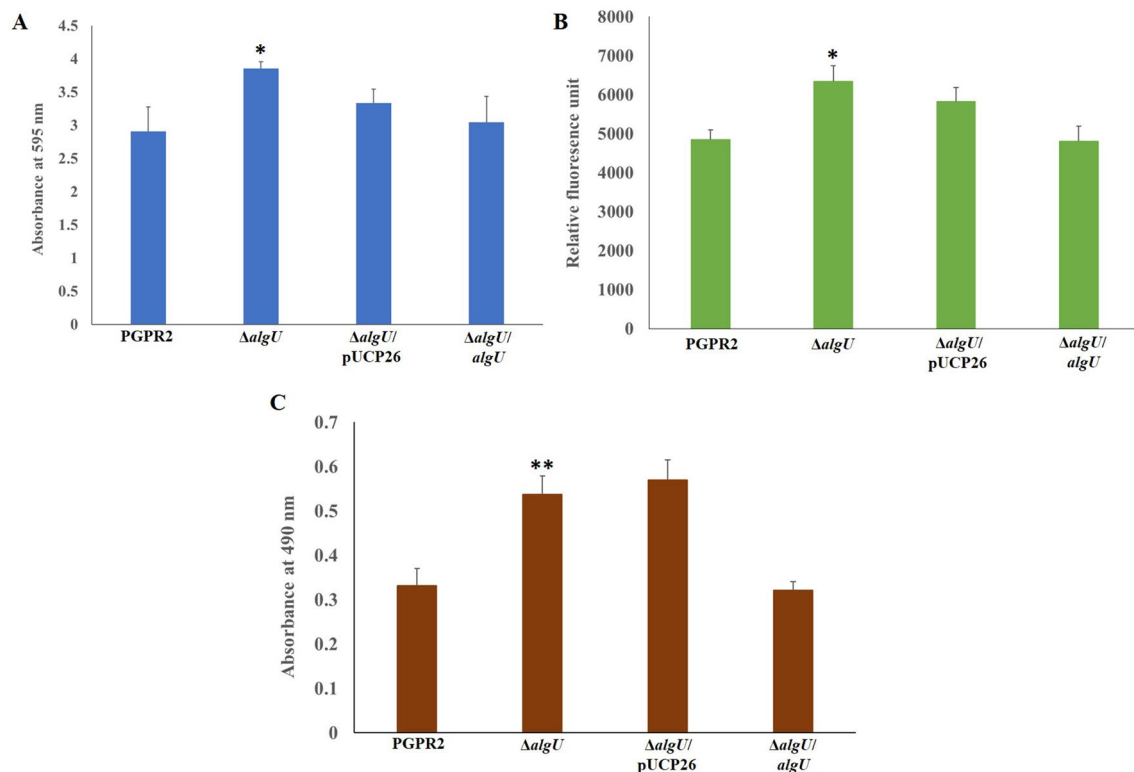


Fig. 5 Biofilm formation by the wild type and the mutant strains on the abiotic surface. **A** Biofilm-forming ability of the PGPR2, the *algU* mutant and the complementation strains on polystyrene plate were assessed using crystal violet staining by measuring the absorbance at 575 nm. * $p=0.012$ **B**. Measurement of cell viability in biofilm using

fluorescein diacetate. The relative difference in fluorescence between the wild type, the mutant and the complementation strains were measured. Increased fluorescence indicated higher biofilm formation. * $p=0.005$ **C** EPS quantification during biofilm formation by the wild type, mutant and complementation strain. ** $p=0.003$

tolerance; this led us to assess the ability of the wild type and the mutant strain to form biofilm on the abiotic surface since the regulatory mechanisms that control stress resistance and biofilm formation have been shown to overlap (Costerton et al. 1994; Hall-Stoodley et al. 2004). Biofilm quantification using crystal violet revealed a significant increase ($p < 0.05$) in biofilm formation by the mutant strain on polystyrene plates, while the complementation strain produced biofilm similar to the parent strain (Fig. 5a). We further quantified the cell viability between the wild type and the mutant strains during biofilm formation by the FDA assay. We observed significantly increased fluorescence in the mutant strains compared with the wild-type strain after 24 h of biofilm formation (Fig. 5b). Similarly, the production of exopolysaccharides was also significantly higher in the mutant strain (Fig. 5c), which is known to play a direct role in biofilm formation (Ghafoor et al. 2011).

We quantified the transcript levels of three genes (*algD*, *pelB*, and *pslA*) involved in forming a matrix during biofilm maturation. The mRNA level of the *algD* gene involved in alginate biosynthesis was down-regulated by 33-fold in the mutant strain, whose expression is directly

regulated by the AlgU sigma factor (Hershberger et al. 1995). In contrast, the transcript levels of genes associated with exopolysaccharide production coded by the *psl* and *pel* genes were up-regulated by 2.9- and 2.6-fold, respectively, in the mutant strain when compared with the wild type during biofilm formation (Fig. 6a). This result further corroborates the increased production of exopolysaccharide by the mutant strain. In addition, we also examined the expression pattern of major fimbrial subunits of the chaperone–usher pathway (cup) (*cupA1*, *cupB1*, and *cupC1*), which play a vital role in adhesion and biofilm formation (Ruer et al. 2007). We observed increased expression of all three genes in the mutant strain with 2.5- to 4.2-fold changes (Fig. 6b).

Discussion

The bacteria of the genus *Pseudomonas* are ubiquitous. They encompass numerous regulatory elements in its genome, which help them to thrive in a wide array of niches by sensing and responding to environmental stimuli. Earlier, we reported the genetic determinants essential for the

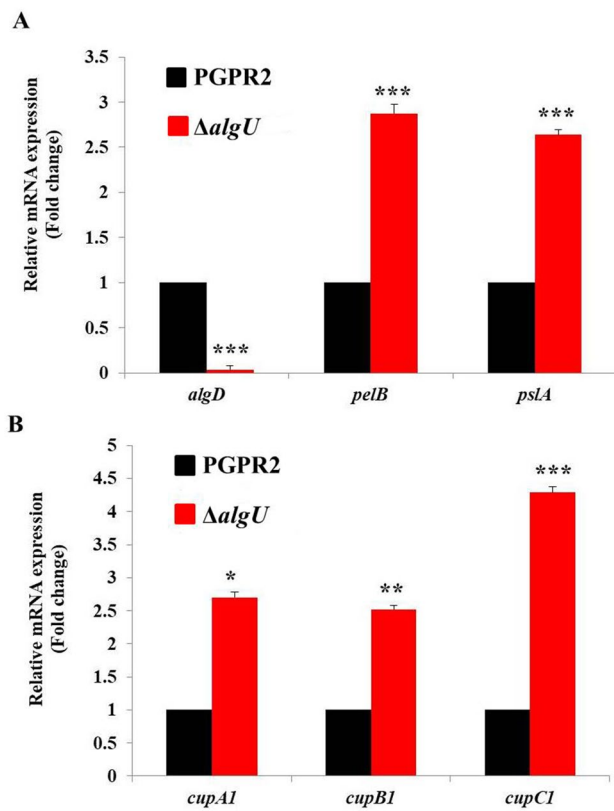


Fig. 6 Expression profile of genes involved in biofilm formation. **A** Effect of *algU* mutation on expression of *algD*, *pelB*, and *pslA* in PGPR2. Significant difference *algD* $p=3.4E-07$; *pelB* $p=3.6E-05$; *pslA* $p=0.0001$. **B** Effect of *algU* mutation on the expression of *cup* fimbrial genes in the PGPR2 strain. Significant difference *cupA1* $p=0.002$; *cupB1* $p=0.0003$; *cupC1* $p=0.0001$. The gene expression value in the wild-type strain for these genes was considered as 1 and calculated the relative change in expression in the mutant strain. Reverse transcriptase quantitative PCR was performed in triplicate, and the data presented are mean of relative change, and the error bar represents the standard deviation

fitness of *P. aeruginosa* PGPR2 during corn root colonization (Sivakumar et al. 2019). During our INSeq screen, we identified transposon integration in gene PGPR2_23995 coding for the AlgU sigma factor to be negatively selected. Transposon integration in the *algU* gene reduced the fitness of PGPR2 by 16.8-fold during root colonization. A previous study reported the involvement of the AlgU sigma factor as a fitness determinant essential for the virulence and colonization of the foliar pathogen *P. syringae* on leaves (Yu et al. 1999, 2014). In this study, we have functionally characterized the role of the AlgU ECF σ in *P. aeruginosa* PGPR2 during corn root colonization. We constructed an isogenic mutant of the *algU* gene by allele exchange mutagenesis. The mutant strain colonized poorly on corn roots both during single inoculation and in a competition assay (Fig. 1) which was in accordance with the previous study (Yu et al. 1999, 2014).

Tolerance to abiotic stresses like osmotic, oxidative, and antibiotic is an essential trait for successful root colonization by rhizobacteria. Plant roots continuously discharge photosynthetically fixed carbon through root exudates leading to increased osmolality (Miller and Wood 1996). Besides serving as a nutrient to the rhizobacteria, root exudates possess a wide array of low molecular weight antimicrobial substances. In addition to regulating the expression of alginate biosynthetic genes, the AlgU ECF σ also modulates other traits like resistance to environmental stresses such as oxidative, osmotic, heat, and antibiotics (Wood and Ohman 2009; Aspedon et al. 2006; Potvin et al. 2008). With this background, we enumerated the degree of resistance of the mutant and the wild-type strain upon induction with different environmental stresses. The growth of the mutant strain was greatly hampered and highly sensitive to osmotic and antibiotic stress, which agreed with previous studies showing a higher expression of the *algU* sigma factor during osmotic and antibiotic stress in *Pseudomonas* species (Schnider-Keel et al. 2001; Wood et al. 2006).

The plant roots constantly release reactive oxygen species (ROS) as a result of aerobic respiration and also serves as a primary defense mechanism of plants against intruders (del Río 2015; Apostol et al. 1989). Earlier, in *Medicago sativa* accumulation of ROS was observed post inoculation with nitrogen-fixing nodulating symbiotic bacteria *Sinorhizobium meliloti* (Santos et al. 2001). A similar effect was observed in plants inoculated with *Burkholderia phytofirmans* PsJN, an endophytic plant growth-promoting rhizobacteria (PGPR) (Bordiec et al. 2011). Mutation in ROS-detoxifying genes hampered the endophytic colonization ability of *Gluconacetobacter diazotrophicus* to rice roots (Alquères et al. 2013). In another study, Verhagen and associates (2010) have shown a transient burst of H_2O_2 by grapevine cells upon inoculation with *P. fluorescens* and *P. aeruginosa*. These studies emphasize the importance of regulating the redox state during the interaction of beneficial bacteria with their cognate host. The mutant strain could resist hydrogen peroxide-induced oxidative stress more efficiently than the wild-type strain.

A previous study reported the role of the AlgU sigma factor in *P. syringae* in combatting oxidative damage. The *algU* mutant strain of *P. syringae* was highly sensitive to oxidative stress induced by paraquat and hydrogen peroxide, which generate reactive oxygen intermediates (Keith and Bender, 1999). This result was in contradiction to our findings. Conversely, in another study, an *algU* mutation in a plant-beneficial biocontrol strain of *P. fluorescens* CHA0 resulted in no significant growth defect in the mutant strain. The mutant strain was less sensitive to paraquat and hydrogen peroxide-induced oxidative stress (Schnider-Keel et al. 2001). This contrasting behavior of the *algU* mutant strains of phytopathogen and plant-beneficial bacteria in resisting

oxidative stress is not clear. One perception is that activating the AlgU sigma factor by reactive-oxygen intermediates helps the phytopathogen withstand the oxidative burst released by the host defense response during infection (Keith and Bender, 1999). On the other hand, this AlgU-mediated defense response is not required for PGPR like *P. fluorescens* CHA0, since they are not often exposed to host defence mechanisms (Schnider-Keel et al. 2001). Further, in *P. fluorescens* strain Pf-5, resistance to oxidative stress is conferred by another sigma factor RpoS (Sarniguet et al. 1995). We speculate similar regulation may be involved in the *algU* mutant of PGPR2 strain, which requires further investigation. Further characterization will provide insight into the regulatory mechanisms governing the enhanced resistance of mutant strain to oxidative stress.

In prokaryotes, post-transcriptional machinery is one of the crucial players in controlling diverse cellular functions by regulating the abundance of specific proteins. One such regulator is RsmA, which modulates gene expression at the post-transcriptional level by binding to the mRNA of target genes. It is a global regulator of gene expression and controls a suite of genes implicated in virulence, biofilm formation, and secondary metabolite production in *P. aeruginosa* (Vakulskas et al. 2015). The transcript level of *rsmA* was down-regulated in the mutant strain, whose expression is positively influenced by the AlgU sigma factor (Stacey and Pritchett 2016). This result led us to measure the expression of genes involved in T3SS. The genetic regulation of T3SS is intricate and governed by several factors, including the AlgU sigma factor and RNA binding protein RsmA (Wu et al. 2004). The expression of genes associated with T3SS was drastically down-regulated in the *algU* mutant, which could be a reason for reduced colonization of the mutant strain on corn roots. The T3SS in PGPR strains plays a crucial role in establishing a mutualistic relationship with the plant host. This notion is supported by several studies, where the inactivation of the T3SS apparatus hampered the root colonization and endophytic lifestyle of PGPR strains (Piromyou et al. 2015; Songwattana et al. 2017; Schmidt et al. 2012; Mavrodi et al. 2011; Rezzonico et al. 2005).

Motility is a crucial and dynamic physiological process in microbes as it facilitates the bacterium to escape from toxic compounds and drives the cells towards nutrient attractants. Our mutant strain displayed augmented swimming motility, evident from increased swimming halos on a semi-solid medium. In pseudomonads, AlgU coordinates the down-regulation of genes associated with motility and flagellar assembly (Martinez-Granero 2012; Muriel et al. 2019). Both plant and animal immune systems recognize peptide

epitopes in flagellin monomer and trigger an immune response (Hajam et al. 2017; Hybiske et al. 2004). The plants are equipped with pattern recognizing receptor that identifies various flagellin epitopes and activates pattern-triggered immunity (PTI) (Zipfel et al. 2004). In *P. syringae* pv. *maculicola* ES4236, the significance of AlgU-mediated signaling cascade involved in de-flagellation was observed. Transposon insertion in the *algW* gene resulted in down-regulation of AlgU and up-regulation of flagellar genes, and a significantly reduced in-plant growth and virulence, which supports our data (Schreiber and Desveaux 2011). The notion that plant associated microbe undergoes de-flagellation was further validated in another study wherein the wild-type strain of *P. syringae* pv. *tomato* DC3000 and its *fliC* mutant displayed similar apoplastic virulence (Bao et al. 2020; Pfeilmeier et al. 2016). The hyper motile phenotype of the *algU* mutant was also attributed to the formation of the second flagellar apparatus (Barahona et al. 2016). Further, the increased expression of the *fliC* and *flhF* genes in the *algU* mutant strain also contributes to the formation of second flagella (Murray and Kazmierczak 2006). These studies provide evidence for the increased activity of flagellar apparatus like flagellin FliC in *algU* mutants results in exposing more peptide epitopes in flagellin monomers which eventually trigger pattern recognition immunity of host plant leading to poor colonization of *algU* mutant. In contrast, the type IV pili-driven twitching motility was significantly hampered in the *algU* mutant strain, due to decreased expression of the *pilZ* gene associated with type IV pili biogenesis. One possible mechanism is the downregulation of the post-transcriptional regulator RsmA in the mutant strain, which controls the expression of an array of genes associated with type IV pili biogenesis (Intile et al. 2014; Burrowes et al. 2006).

Conclusion

We have demonstrated that the AlgU sigma factor contributes to the fitness of PGPR2 during root colonization. Our data shed light on the key genetic regulations orchestrated by the AlgU sigma factor in PGPR. The schematic representation of the *algU*-mediated gene expression is shown in Fig. 7. The regulation of gene expression by the AlgU sigma factor is multifaceted. Understanding these signaling mechanisms would provide novel insights into functions governed by the sigma factor besides the regulation of alginate biosynthesis.

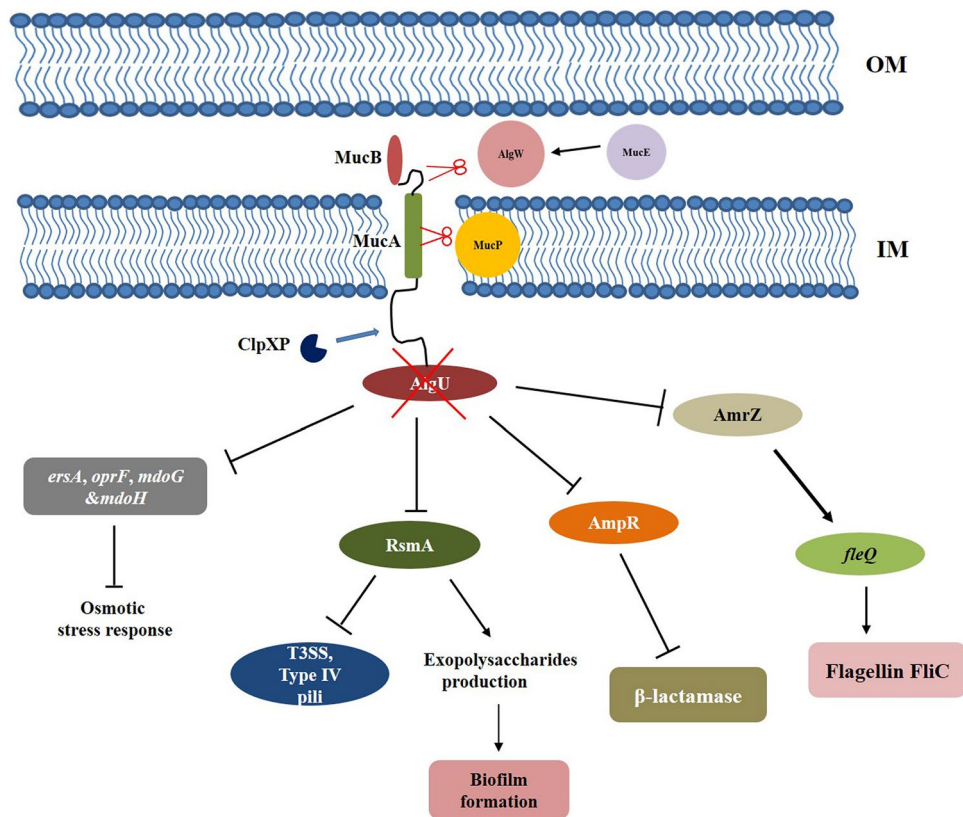


Fig. 7 A model for *algU*-mediated regulation of gene expression in PGPR2. Inactivation of *algU* sigma factor hampers expression of sRNA (*ersA*), the major outer membrane protein (*oprF*), and two genes forming osmoregulated periplasmic glucans (*mdoH* and *mdoG*). These genes play a crucial role in osmotic stress tolerance and regulated by the *algU* sigma factor (Wood and Ohman 2009; Ferrara et al. 2015; Park et al. 2014). Similarly, *algU* mutant display reduced expression of β -lactamase, which confers resistance against antibiotic stress. The *algU* sigma factor positively influences the production of β -lactamase through an *ampR* regulator, which is under the control

of *algU* (Balasubramanian et al. 2011). Mutation in the *algU* sigma factor also impeded the expression of global regulator *rsmA*, which is positively modulated by *algU* (Stacey and Pritchett 2016). The post-transcriptional regulator *rsmA* positively regulates T3SS and type IV pili biosynthesis (Intile et al. 2014) and negatively regulates genes implicated with biofilm formation (Brenic and Lory 2009). Inactivation of *algU* leads to decrease expression of *amrZ* which activates *fleQ* leading to increased synthesis of flagella apparatus including flagellin FliC (Martinez-Granero 2012). OM Outer membrane; IM Inner membrane

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Authors contributions JR, PG, and RS: Conceived and designed the work; RS: Performed the experiments; RS and JR: Analyzed and interpreted data; RS, PG, and JR: Wrote the manuscript.

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Declarations

Conflict of interest RS declares that he/she has no conflict of interest. PG declares that he/she has no conflict of interest. JR declares that he/she has no conflict of interest.

Ethical approval This article does not contain any studies with human participants or animals performed by any of the authors.

References

- Alm RA, Mattick JS (1995) Identification of a gene, pilV, required for type 4 fimbrial biogenesis in *Pseudomonas aeruginosa*, whose product possesses a pre-pilin-like leader sequence. Mol Microbiol 16:485–496. <https://doi.org/10.1111/j.1365-2958.1995.tb02413.x>
- Alquéres S, Meneses C, Rouws L, Rothballe M, Baldani I, Schmid M, Hartmann A (2013) The bacterial superoxide dismutase and

- glutathione reductase are crucial for endophytic colonization of rice roots by *Gluconacetobacter diazotrophicus* PAL5. *Mol Plant Microbe Interact* 26:937–945. <https://doi.org/10.1094/MPMI-12-12-0286-R>
- Apostol I, Heinsteins PF, Low PS (1989) Rapid stimulation of an oxidative burst during elicitation of cultured plant cells: role in defense and signal transduction. *Plant Physiol* 90:109–116. <https://doi.org/10.1104/pp.90.1.109>
- Aspedon A, Palmer K, Whiteley M (2006) Microarray analysis of the osmotic stress response in *Pseudomonas aeruginosa*. *J Bacteriol* 188:2721–2725. <https://doi.org/10.1128/JB.188.7.2721-2725.2006>
- Balasubramanian D, Kong KF, Jayawardena SR, Leal SM, Sautter RT, Mathee K (2011) Co-regulation of beta-lactam resistance, alginate production and quorum sensing in *Pseudomonas aeruginosa*. *J Med Microbiol* 60:147–156. <https://doi.org/10.1099/jmm.0.021600-0>
- Bao Z, Wei HL, Ma X, Swingle B (2020) *Pseudomonas syringae* AlgU downregulates flagellin gene expression, helping evade plant immunity. *J Bacteriol* 202:e00418-e419. <https://doi.org/10.1128/JB.00418-19>
- Barahona E, Navazo A, Garrido-Sanz D et al (2016) *Pseudomonas fluorescens* F113 can produce a second flagellar apparatus, which is important for plant root colonization. *Front Microbiol* 7:1471. <https://doi.org/10.3389/fmicb.2016.01471>
- Bordes P, Lavatine L, Phok K, Barriot R, Boulanger A, Castanie Cornet MP, Dejean G, Lauber E, Becker A, Arlat M, Gutierrez C (2011) Insights into the extracytoplasmic stress response of *Xanthomonas campestris* pv. *campestris*: Role and regulation of σ E-dependent activity. *J Bacteriol* 193:246–264. <https://doi.org/10.1128/JB.00884-10>
- Bordiec S, Paquis S, Lacroix H, Dhondt S, Ait Barka E, Kauffmann S, Jeandet P, Mazeyrat-Gourbeyre F, Clément C, Baillieul F, Dorey S (2011) Comparative analysis of defence responses induced by the endophytic plant growth-promoting rhizobacterium *Burkholderia phytofirmans* strain PsJN and the non-host bacterium *Pseudomonas syringae* pv. *pisi* in grapevine cell suspensions. *J Exp Bot* 62:595–603. <https://doi.org/10.1093/jxb/erq291>
- Brencic A, Lory S (2009) Determination of the regulon and identification of novel mRNA targets of *Pseudomonas aeruginosa* RsmA. *Mol Microbiol* 72:612–632. <https://doi.org/10.1111/j.1365-2958.2009.06670.x>
- Burrowes E, Bayse C, Adams C, O’Gara F (2006) Influence of the regulatory protein RsmA on cellular functions in *Pseudomonas aeruginosa* PAO1, as revealed by transcriptome analysis. *Microbiol* 152:405–418. <https://doi.org/10.1099/mic.0.28324-0>
- Cheng CY, Shieh SY, Hsu CC, Yang MT (2008) Characterization and transcriptional analysis of an ECF sigma factor from *Xanthomonas campestris* pv. *campestris*. *FEMS Microbiol Lett* 289:250–257. <https://doi.org/10.1111/j.1574-6968.2008.01392.x>
- Chevalier S, Bouffartigues E, Bazire A et al (2019) Extracytoplasmic function sigma factors in *Pseudomonas aeruginosa*. *BBA Gene Regul Mech* 1862:706–721. <https://doi.org/10.1016/j.bbagr.2018.04.008>
- Costerton JW, Lewandowski Z, DeBeer D, Caldwell D, Korber D, James G (1994) Biofilms, the customized microniche. *J Bacteriol* 176:2137–2142. <https://doi.org/10.1128/jb.176.8.2137-2142.1994>
- da Silva Neto JF, Koide T, Gomes SL, Marques MV (2007) The single extracytoplasmic-function sigma factor of *Xylella fastidiosa* is involved in the heat shock response and presents an unusual regulatory mechanism. *J Bacteriol* 189:551–560. <https://doi.org/10.1128/JB.00986-06>
- Damron FH, Goldberg JB (2012) Proteolytic regulation of alginate overproduction in *Pseudomonas aeruginosa*. *Mol Microbiol* 84:95–607. <https://doi.org/10.1111/j.1365-2958.2012.08049.x>
- de Lorenzo V, Herrero M, Jakubzik U, Timmis KN (1990) Mini-Tn5 transposon derivatives for insertion mutagenesis, promoter probing, and chromosomal insertion of cloned DNA in Gram-negative Eubacteria. *J Bacteriol* 172:6568–6572. <https://doi.org/10.1128/jb.172.11.6568-6572.1990>
- del Río LA (2015) ROS and RNS in plant physiology: an overview. *J Exp Bot* 66:2827–2837. <https://doi.org/10.1093/jxb/erv099>
- Fan B, Carvalhais LC, Becker A, Fedoseyenko D, von Wirén N, Borriss R (2012) Transcriptomic profiling of *Bacillus amyloliquefaciens* FZB42 in response to maize root exudates. *BMC Microbiol* 12:116. <https://doi.org/10.1186/1471-2180-12-116>
- Ferrara S, Carloni S, Fulco R, Falcone M, Macchi R, Bertoni G (2015) Post-transcriptional regulation of the virulence-associated enzyme AlgC by the σ 22-dependent small RNA ErsA of *Pseudomonas aeruginosa*. *Environ Microbiol* 17:199–214. <https://doi.org/10.1111/1462-2920.12590>
- Firoved AM, Boucher JC, Deretic V (2002) Global genomic analysis of AlgU (σ E)-dependent promoters (sigmulon) in *Pseudomonas aeruginosa* and implications for inflammatory processes in cystic fibrosis. *J Bacteriol* 184:1057–1064. <https://doi.org/10.1128/jb.184.4.1057-1064.2002>
- Ghafoor A, Hay ID, Rehm BH (2011) Role of exopolysaccharides in *Pseudomonas aeruginosa* biofilm formation and architecture. *Appl Environ Microbiol* 77:5238–5246. <https://doi.org/10.1128/AEM.00637-11>
- Hajam IA, Dar PA, Shahnawaz I, Jaume JC, Lee JH (2017) Bacterial flagellin—a potent immunomodulatory agent. *Exp Mol Med* 49:e373. <https://doi.org/10.1038/emm.2017.172>
- Hall-Stoodley L, Costerton JW, Stoodley P (2004) Bacterial biofilms: from the natural environment to infectious diseases. *Nat Rev Microbiol* 2:95–108. <https://doi.org/10.1038/nrmicro821>
- Helmann JD (2002) The extracytoplasmic function (ECF) sigma factors. *Adv Microb Physiol* 46:47–110. [https://doi.org/10.1016/S0065-2911\(02\)46002-X](https://doi.org/10.1016/S0065-2911(02)46002-X)
- Hershberger CD, Ye RW, Parsek MR, Xie ZD, Chakrabarty AM (1995) The *algT* (*algU*) gene of *Pseudomonas aeruginosa*, a key regulator involved in alginate biosynthesis, encodes an alternative sigma factor (sigma E). *Proc Natl Acad Sci USA* 92:7941–7945. <https://doi.org/10.1073/pnas.92.17.7941>
- Hybiske K, Ichikawa JK, Huang V, Lory SJ, Machen TE (2004) Cystic fibrosis airway epithelial cell polarity and bacterial flagellin determine host response to *Pseudomonas aeruginosa*. *Cell Microbiol* 6:49–63. <https://doi.org/10.1046/j.1462-5822.2003.00342.x>
- Illakkiam D, Ponraj P, Shankar M, Muthusubramanian S, Rajendhran J, Gunasekaran P (2013) Identification and structure elucidation of a novel antifungal compound produced by *Pseudomonas aeruginosa* PGPR2 against *Macrophomina phaseolina*. *Appl Biochem Biotechnol* 171:2176–2185. <https://doi.org/10.1007/s12010-013-0469-7>
- Intile PJ, Diaz MR, Urbanowski ML, Wolfgang MC, Yahr TL (2014) The AlgZR two-component system recalibrates the RsmAYZ posttranscriptional regulatory system to inhibit expression of the *Pseudomonas aeruginosa* type III secretion system. *J Bacteriol* 196:357–366. <https://doi.org/10.1128/JB.01199-13>
- Keith LMW, Bender CL (1999) AlgT (σ 22) controls alginate production and tolerance to environmental stress in *Pseudomonas syringae*. *J Bacteriol* 181:7176–7184. <https://doi.org/10.1128/JB.181.23.7176-7184.1999>
- Livak KJ, Schmittgen TD (2001) Analysis of relative gene expression data using real-time quantitative PCR and the $2^{-\Delta\Delta CT}$ method. *Methods* 25:402–408. <https://doi.org/10.1006/meth.2001.1262>
- Lyczak JB, Cannon CL, Pier GB (2002) Lung infections associated with cystic fibrosis. *Clin Microbiol Rev* 15:194–222. <https://doi.org/10.1128/cmr.15.2.194-222.2002>
- Martin DW, Holloway BW, Deretic V (1993) Characterization of a locus determining the mucoid status of *Pseudomonas aeruginosa*:

- AlgU shows sequence similarities with a *Bacillus* sigma factor. J Bacteriol 175:1153–1164. <https://doi.org/10.1128/jb.175.4.1153-1164.1993>
- Martínez-Bueno MA, Tobes R, Rey M, Ramos JL (2002) Detection of multiple extracytoplasmic function (ECF) sigma factors in the genome of *Pseudomonas putida* KT2440 and their counterparts in *Pseudomonas aeruginosa* PA01. Environ Microbiol 4:842–855. <https://doi.org/10.1046/j.1462-2920.2002.00371.x>
- Martínez-Granero F, Navazo A, Barahona E, Redondo-Nieto M, Rivilla R, Martín M (2012) The Gac-Rsm and SadB signal transduction pathways converge on AlgU to downregulate motility in *Pseudomonas fluorescens*. PLoS ONE 7:e31765. <https://doi.org/10.1371/journal.pone.0031765>
- Mavrodi DV, Joe A, Mavrodi OV et al (2011) Structural and functional analysis of the type III secretion system from *Pseudomonas fluorescens* Q8r1–96. J Bacteriol 193:177–189. <https://doi.org/10.1128/JB.00895-10>
- Miller KJ, Wood JM (1996) Osmoadaptation by rhizosphere bacteria. Annu Rev Microbiol 50:101–136. <https://doi.org/10.1146/annurev.micro.50.1.101>
- Murray TS, Kazmierczak BI (2006) FlhF is required for swimming and swarming in *Pseudomonas aeruginosa*. J Bacteriol 188:6995–7004. <https://doi.org/10.1128/JB.00790-06>
- Muriel C, Blanco-Romero E, Trampari E, Arrebola E, Durán D, Redondo-Nieto M, Malone JG, Martín M, Rivilla R (2019) The diguanylate cyclase AdrA regulates flagellar biosynthesis in *Pseudomonas fluorescens* F113 through SadB. Sci Rep 9:8096. <https://doi.org/10.1038/s41598-019-44554-z>
- Müsken M, Di Fiore S, Römling U, Häussler S (2010) A 96-well-plate-based optical method for the quantitative and qualitative evaluation of *Pseudomonas aeruginosa* biofilm formation and its application to susceptibility testing. Nat Protoc 5:1460. <https://doi.org/10.1038/nprot.2010.110>
- Nelson KE, Weinelt C, Paulsen IT et al (2002) Complete genome sequence and comparative analysis of the metabolically versatile *Pseudomonas putida* KT2440. Environ Microbiol 4:799–808. <https://doi.org/10.1046/j.1462-2920.2002.00366.x>
- Nithya C, Devi MG, Karutha Pandian S (2011) A novel compound from the marine bacterium *Bacillus pumilus* S6–15 inhibits biofilm formation in gram-positive and gram-negative species. Biofouling 27:519–528. <https://doi.org/10.1080/08927014.2011.586127>
- Paget MS, Helmann JD (2003) The $\sigma 70$ family of sigma factors. Genome Biol 4:203. <https://doi.org/10.1186/gb-2003-4-1-203>
- Park SH, Bao Z, Butcher BG et al (2014) Analysis of the small RNA *spf* in the plant pathogen *Pseudomonas syringae* pv. tomato strain DC3000. Microbiol 160:941–953. <https://doi.org/10.1099/mic.0.076497-0>
- Peeters E, Nelis HJ, Coenye T (2008) Comparison of multiple methods for quantification of microbial biofilms grown in microtiter plates. J Microbiol Meth 72:157–165. <https://doi.org/10.1016/j.mimet.2007.11.010>
- Pfeilmeier S, Saur IM, Rathjen JP, Zipfel C, Malone JG (2016) High levels of cyclic-di-GMP in plant-associated *Pseudomonas* correlate with evasion of plant immunity. Mol Plant Pathol 17:521–531. <https://doi.org/10.1111/mp.12297>
- Piromy P, Songwattana P, Greetatorn T et al (2015) The type III secretion system (T3SS) is a determinant for rice-endophyte colonization by non-photosynthetic *Bradyrhizobium*. Microbes Environ 30:291–300. <https://doi.org/10.1264/jsme2.ME15080>
- Potvin E, Sanschagrin F, Levesque R (2008) Sigma factors in *Pseudomonas aeruginosa*. FEMS Microbiol Rev 32:38–55. <https://doi.org/10.1111/j.1574-6976.2007.00092.x>
- Qiu D, Eisinger VM, Rowen DW, Hongwei DY (2007) Regulated proteolysis controls mucoid conversion in *Pseudomonas aeruginosa*. P Natl Acad Sci USA 104:8107–8112. <https://doi.org/10.1073/pnas.0702660104>
- Rainey PB (1999) Adaptation of *Pseudomonas fluorescens* to the plant rhizosphere. Environ Microbiol 1:243–257. <https://doi.org/10.1046/j.1462-2920.1999.00040.x>
- Ramsey DM, Wozniak DJ (2005) Understanding the control of *Pseudomonas aeruginosa* alginate synthesis and the prospects for management of chronic infections in cystic fibrosis. Mol Microbiol 56:309–322. <https://doi.org/10.1111/j.1365-2958.2005.04552.x>
- Rezzonico F, Binder C, Défago G, Moëne-Loccoz Y (2005) The type III secretion system of biocontrol *Pseudomonas fluorescens* KD targets the phytopathogenic Chromista *Pythium ultimum* and promotes cucumber protection. Mol Plant Microbe 18:991–1001. <https://doi.org/10.1094/MPMI-18-0991>
- Ruer S, Stender S, Filloux A, de Bentzmann S (2007) Assembly of fimbrial structures in *Pseudomonas aeruginosa*: functionality and specificity of chaperone-usher machineries. J Bacteriol 189:3547–3555. <https://doi.org/10.1128/JB.00093-07>
- Sambrook J, Fritsch EF, Maniatis T (1989) Molecular cloning: a laboratory manual, 2nd edn. Cold Spring Harbor Laboratory Press, Cold Spring Harbor
- Santos R, Hérouart D, Sigaud S, Touati D, Puppo A (2001) Oxidative burst in alfalfa *Sinorhizobium meliloti* symbiotic interaction. Mol Plant-Microbe 14:86–89. <https://doi.org/10.1094/MPMI.2001.14.1.86>
- Sarniguet A, Kraus J, Henkels MD, Muehlchen AM, Loper JE (1995) The sigma factor sigma_S affects antibiotic production and biological control activity of *Pseudomonas fluorescens* Pf-5. Proc Natl Acad Sci USA 92:12255–12259. <https://doi.org/10.1073/pnas.92.26.12255>
- Schmidt MA, Balsanelli E, Faoro H et al (2012) The type III secretion system is necessary for the development of a pathogenic and endophytic interaction between *Herbaspirillum rubrisubalbicans* and *Poaceae*. BMC Microbiol 12:98. <https://doi.org/10.1186/1471-2180-12-98>
- Schnider-Keel U, Lejbølle KB, Baehler E, Haas D, Keel C (2001) The Sigma Factor AlgU (AlgT) controls exopolysaccharide production and tolerance towards desiccation and osmotic stress in the biocontrol agent *Pseudomonas fluorescens* CHA0. Appl Environ Microbiol 67:5683–5693. <https://doi.org/10.1128/AEM.67.12.5683-5693.2001>
- Schreiber KJ, Desveaux D (2011) AlgW regulates multiple *Pseudomonas syringae* virulence strategies. Mol Microbiol 80:364–377. <https://doi.org/10.1111/j.1365-2958.2011.07571.x>
- Schurr MJ, Martin DW, Mudd MH, Deretic V (1994) Gene cluster controlling conversion to alginate-overproducing phenotype in *Pseudomonas aeruginosa*: functional analysis in a heterologous host and role in the instability of mucoidy. J Bacteriol 176:3375–3382. <https://doi.org/10.1128/jb.176.11.3375-3382.1994>
- Schurr MJ, Yu H, Martínez-Salazar JM, Boucher JC, Deretic V (1996) Control of AlgU, a member of the sigma E-like family of stress sigma factors, by the negative regulators MucA and MucB and *Pseudomonas aeruginosa* conversion to mucoidy in cystic fibrosis. J Bacteriol 178:4997–5004. <https://doi.org/10.1128/jb.178.16.4997-5004.1996>
- Semmler AB, Whitchurch CB, Mattick JS (1999) A re-examination of twitching motility in *Pseudomonas aeruginosa*. Microbiology 145:2863–2873. <https://doi.org/10.1099/00221287-145-10-2863>
- Shi XY, Dumenyo CK, Hernandez-Martinez R, Azad H, Cooksey DA (2007) Characterization of regulatory pathways in *Xylella fastidiosa*: Genes and phenotypes controlled by algU. Appl Environ Microbiol 73:6748–6756. <https://doi.org/10.1128/AEM.01232-07>
- Shidore T, Dinse T, Öhrlein J, Becker A, Reinhold-Hurek B (2012) Transcriptomic analysis of responses to exudates reveal genes required for rhizosphere competence of the endophyte *Azoarcus*

- sp. strain BH72. *Enviro Microbiol* 14:2775–2787. <https://doi.org/10.1111/j.1462-2920.2012.02777.x>
- Sivakumar R, Ranjani J, Vishnu US et al (2019) Evaluation of INSeq to identify genes essential for *Pseudomonas aeruginosa* PGPR2 corn root colonization. *G3. Genes Genom Genet* 9:651–661. <https://doi.org/10.1534/g3.118.200928>
- Songwattana P, Noisangiam R, Teamtisong K et al (2017) Type 3 secretion system (T3SS) of *Bradyrhizobium* sp. DOA9 and its roles in legume symbiosis and rice endophytic association. *Front Microbiol* 8:1810. doi:<https://doi.org/10.3389/fmicb.2017.01810>
- Stacey SD, Pritchett CL (2016) *Pseudomonas aeruginosa* AlgU contributes to post-transcriptional activity by increasing *rsmA* expression in a mucA22 strain. *J Bacteriol* 198:1812–1826. <https://doi.org/10.1128/JB.00133-16>
- Staron A, Sofia HJ, Dietrich S, Ulrich LE, Liesegang H, Mascher T (2009) The third pillar of bacterial signal transduction: classification of the extracytoplasmic function (ECF) σ factor protein family. *Mol Microbiol* 74:557–581. <https://doi.org/10.1111/j.1365-2958.2009.06870.x>
- Stover CK, Pham XQ, Erwin AL et al (2000) Complete genome sequence of *Pseudomonas aeruginosa* PAO1, an opportunistic pathogen. *Nature* 406:959. <https://doi.org/10.1038/35023079>
- Vakulskas CA, Potts AH, Babitzke P, Ahmer BM, Romeo T (2015) Regulation of bacterial virulence by Csr (Rsm) systems. *Microbiol Mol Biol Rev* 79:193–224. <https://doi.org/10.1128/MMBR.00052-14>
- Verhagen BWM, Trostel-Aziz P, Couderchet M, Höfte M, Aziz A (2010) *Pseudomonas* spp. induced systemic resistance to *Botrytis cinerea* is associated with induction and priming of defence responses in grapevine. *J Exp Bot* 61:249–260. <https://doi.org/10.1093/jxb/erp295>
- West SEH, Schweizer HP, Dall C, Sample AK, Runyen-Janecky LJ (1994) Construction of improved *Escherichia-Pseudomonas* shuttle vectors derived from pUC18/19 and sequence of the region required for their replication in *Pseudomonas aeruginosa*. *Gene* 148:81–86. [https://doi.org/10.1016/0378-1119\(94\)90237-2](https://doi.org/10.1016/0378-1119(94)90237-2)
- Wood LF, Leech AJ, Ohman DE (2006) Cell wall-inhibitory antibiotics activate the alginate biosynthesis operon in *Pseudomonas aeruginosa*: roles of σ 22 (AlgT) and the AlgW and Prc proteases. *Mol Microbiol* 62:412–426. <https://doi.org/10.1111/j.1365-2958.2006.05390.x>
- Wood LF, Ohman DE (2009) Use of cell wall stress to characterize σ 22 (AlgT/U) activation by regulated proteolysis and its regulon in *Pseudomonas aeruginosa*. *Mol Microbiol* 72:183–201. <https://doi.org/10.1111/j.1365-2958.2009.06635.x>
- Wu W, Badrane H, Arora S, Baker HV, Jin S (2004) MucA-mediated coordination of type III secretion and alginate synthesis in *Pseudomonas aeruginosa*. *J Bacteriol* 186:7575–7585. <https://doi.org/10.1128/JB.186.22.7575-7585.2004>
- Xie ZD, Hershberger CD, Shankar S, Ye RW, Chakrabarty AM (1996) Sigma factor-anti-sigma factor interaction in alginate synthesis: inhibition of AlgT by MucA. *J Bacteriol* 178:4990–4996. <https://doi.org/10.1128/jb.178.16.4990-4996.1996>
- Yang LY, Yang LC, Gan YL, Wang L, Zhao WZ, He YQ, Jiang W, Jiang BL, Tang JL (2018) Systematic functional analysis of sigma (s) factors in the phytopathogen *Xanthomonas campestris* reveals novel roles in the regulation of virulence and viability. *Front Microbiol* 9:1749. <https://doi.org/10.3389/fmicb.2018.01749>
- Yi Y, de Jong A, Frenzel E, Kuipers OP (2017) Comparative transcriptomics of *Bacillus mycoides* strains in response to potato-root exudates reveals different genetic adaptation of endophytic and soil isolates. *Front Microbiol* 8:1487. <https://doi.org/10.3389/fmicb.2017.01487>
- Yu J, Peñaloza-Vázquez A, Chakrabarty AM, Bender CL (1999) Involvement of the exopolysaccharide alginate in the virulence and epiphytic fitness of *Pseudomonas syringae* pv. *syringae*. *Mol Microbiol* 33:712–720. <https://doi.org/10.1046/j.1365-2958.1999.01516.x>
- Yu X, Lund SP, Greenwald JW et al (2014) Transcriptional analysis of the global regulatory networks active in *Pseudomonas syringae* during leaf colonization. *Mbio* 5:e01683-e1714. <https://doi.org/10.1128/mBio.01683-14>
- Zhang N, Yang D, Wang D et al (2015) Whole transcriptomic analysis of the plant-beneficial rhizobacterium *Bacillus amyloliquefaciens* SQR9 during enhanced biofilm formation regulated by maize root exudates. *BMC Genomics* 16:685. <https://doi.org/10.1186/s12864-015-1825-5>
- Zipfel C, Robatzek S, Navarro L, Oakeley EJ, Jones JD, Felix G, Boller T (2004) Bacterial disease resistance in *Arabidopsis* through flagellin perception. *Nature* 428:764–767

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